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SPECIFICATION FOR APPROVAL

DATE：

CUSTOMER：

PART NAME： General Purpose Lead Free &. Halogen Free Chip Resistors

CUSTOMER'S DWG. NO.：

CUSTOMER'S PART NO.：

PDC PART NO.： **FCF-G SERIES APPROVED**

DESCRIPTION.：

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ACTION / RESULT	" V "	CUSTOMER'S SIGNATURE	NOTE
FULL APPROVED			
CONDITIONAL APPROVED			
REJECTED			

OUR ACTION	SIGNATURE
PREPARED By	Jenny Tseng
CHECKED By	Tony Chou
APPROVED By	Byron Tsai

CUSTOMER SIGNATURE FOR ACCEPTANCE

PDC

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FCF Green series.
RoHS Exemption Free (Pb≦100ppm)
Thick-film Lead Free Chip Resistors

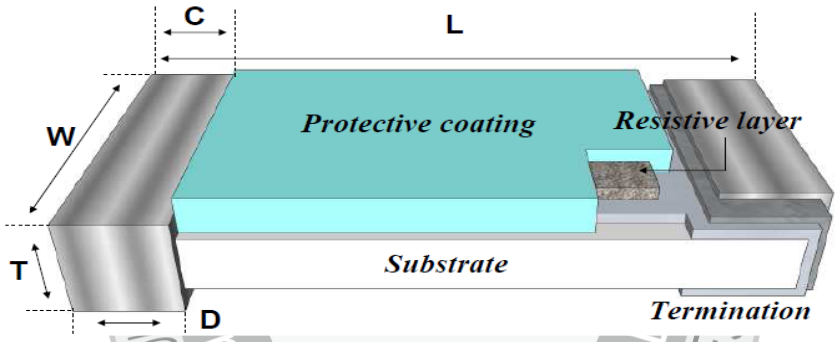
1. Features

- Small size and light weight.
- Suitable for lead free soldering.
- Compatible with wave and reflow soldering.
- RoHS compliant & Halogen free.
- Lead content below100ppm.

2. Applications

- Mobile phone.
- Digital meter, Consumer electronics, M/B.
- Portable electronics devices.

3.Dimension and Construction



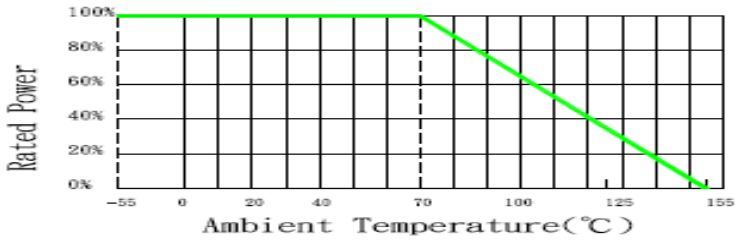
Unit：mm

Type	L	W	C	D	T
FCF02	1.00±0.05	0.50±0.05	0.20±0.10	0.25±0.10	0.35±0.05
FCF03	1.60±0.10	0.80±0.10	0.30±0.20	0.30±0.20	0.45±0.10
FCF05	2.00±0.10	1.25±0.10	0.40±0.20	0.40±0.20	0.50±0.10
FCF06	3.10±0.10	1.60±0.10	0.50±0.20	0.50±0.25	0.55±0.10
FCF12	3.10±0.10	2.60±0.15	0.50±0.25	0.50±0.25	0.55±0.10
FCF20	5.00±0.20	2.50±0.20	0.65±0.25	0.60±0.25	0.55±0.10
FCF25	6.40±0.20	3.20±0.20	0.65±0.25	0.90±0.25	0.60±0.10
FCF18	3.05±0.15	4.60±0.20	0.45±0.25	0.50±0.25	0.55±0.10

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4.Power Derating Curve

Operating Temperature Range: -55 to +155 deg.C



5.Rating

Type	Size	Power Rating at 70℃	Max. RCWV	Max. Overload Voltage	Resistance Tolerance	Temperature Coefficient (ppm/℃)	Resistance Range		Standard Resistance Values
							Min.	Max.	
FCF02	0402	1/16W	50V	100V	±1%(F)	-300/+500	1 Ω	10 Ω	E24 E96
						±100	10.2 Ω	976 KΩ	
						±300	1 MΩ	10 MΩ	
					±5%(J)	-300/+500	1 Ω	10 Ω	E24 Jumper
						±200	11 Ω	910 KΩ	
						±300	1 MΩ	10 MΩ	
FCF03	0603	1/10W	50V	100V	±1%(F)	-300/+500	1 Ω	10 Ω	E24 E96
						±100	10.2 Ω	976 KΩ	
						±200	1 MΩ	10 MΩ	
					±5%(J)	-300/+500	1 Ω	10 Ω	E24 Jumper
						±200	11 Ω	910 KΩ	
						±200	1 MΩ	10 MΩ	
FCF05	0805	1/8W	150V	300V	±1%(F)	-300/+500	1 Ω	10 Ω	E24 E96
						±100	10.2 Ω	976 KΩ	
						±200	1 MΩ	10 MΩ	
					±5%(J)	-300/+500	1 Ω	10 Ω	E24 Jumper
						±200	11 Ω	910 KΩ	
						±200	1 MΩ	10 MΩ	

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FCF Green series.
RoHS Exemption Free (Pb≦100ppm)
Thick-film Lead Free Chip Resistors

Type	Size	Power Rating at 70℃	Max. RCWV	Max. Overload Voltage	Resistance Tolerance	Temperature Coefficient (ppm/℃)	Resistance Range		Standard Resistance Values
							Min.	Max.	
FCF06	1206	1/4W	200V	400V	±1%(F)	-300/+500	1 Ω	10 Ω	E24 E96
						±100	10.2 Ω	976 KΩ	
						±200	1 MΩ	10 MΩ	
					±5%(J)	-300/+500	1 Ω	10 Ω	E24 Jumper
						±200	11 Ω	910 KΩ	
						±200	1 MΩ	10 MΩ	
FCF12	1210	1/3W	200V	400V	±1%(F)	-300/+500	1 Ω	10 Ω	E24 E96
						±100	10.2 Ω	976 KΩ	
						±200	1 MΩ	10 MΩ	
					±5%(J)	-300/+500	1 Ω	10 Ω	E24 Jumper
						±200	11 Ω	910 KΩ	
						±200	1 MΩ	10 MΩ	
FCF20	2010	1/2W	200V	400V	±1%(F)	±100	1 Ω	10 Ω	E24 E96
						±200	10.2 Ω	10 MΩ	
					±5%(J)	±100	1 Ω	10 Ω	E24 Jumper
						±200	11 Ω	10 MΩ	
FCF25	2512	1W	250V	500V	±1%(F)	±100	1 Ω	10 Ω	E24 E96
						±200	10.2 Ω	10 MΩ	
					±5%(J)	±100	1 Ω	10 Ω	E24 Jumper
						±200	11 Ω	10 MΩ	
FCF18	1218	1W	200V	400V	±1%(F)	±100	1 Ω	10 Ω	E24 E96
						±200	10.2 Ω	10 MΩ	
					±5%(J)	±100	1 Ω	10 Ω	E24 Jumper
						±200	11 Ω	10 MΩ	

Note：RCWV＝$(P\times R)^{1/2}$ or Max. RCWV listed above, whichever is lower.

RCWV：Working Voltage (V)，P：Rated Power (W)，R：Resistance Value (Ω)

Jumper：Max. 50mΩ.

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6.Part Number

Type	Size	Tolerance	Packing	Watt	R Value	TCR	Control Code
FCF	**02** :0402 **03** :0603 **05** :0805 **06** :1206 **20** :2010 **25** :2512 **18** :1218	**F** :±1% **J** :±5%	**Paper Tape：** **0402.** **V** : 10Kpcs **0603.0805.** **1206.1210.** **T** : 5Kpcs **V** : 10Kpcs **W** : 20Kpcs **Plastic Tape：** **2010.2512** **P** : 4Kpcs **1218** **Q** : 3Kpcs	**-**: As Rating Info.	**XXXX** **XXX** **1%:** 4 digits **5%:** 3 digits	**-**: As Rating Info.	**G** : Green series.

Example：

FCF05FT-1001-G

➔0805 size, tolerance 1%, paper tape, 1/8W, 1 KΩ, Green series.

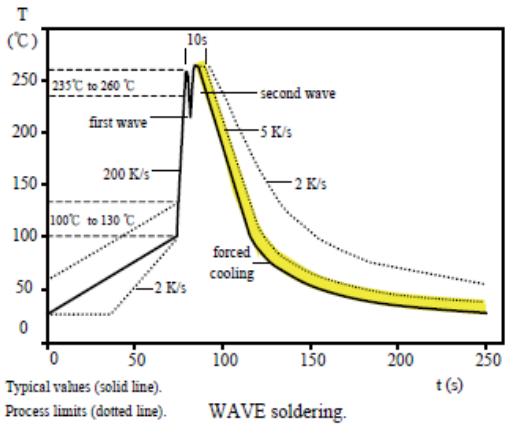
FCF03JT-203_-G

➔0603 size, tolerance 5%, paper tape, 1/10W, 20 KΩ, Green series.

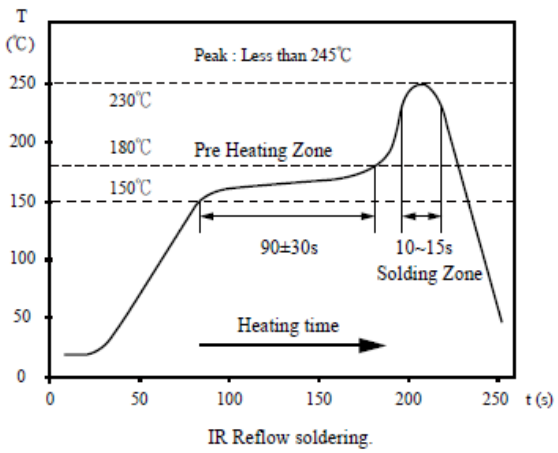
7.Marking/Soldering

※ All sizes are no marking.

Soldering Reference：



WAVE soldering.



IR Reflow soldering.

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8. Reliability Performance

Test Item	Specification	Test Method
DC Resistance	F:±1% J:±5%	**IEC 60115-1 / JIS C 5201-1 , Clause 4.5** Measure the resistance Value.
Resistance to Solder Heat	2010/2512/1218： ΔR≦±(1%＋0.05Ω) Other size： F:ΔR≦±(0.5%＋0.1Ω) J:ΔR≦±(1%＋0.1Ω)	**IEC 60115-1, Clause 4.18** Solder dipping @ 260℃±5℃ for 10sec.±1sec.
Solder Ability	Over 95% of termination must be covered with solder.	**IEC 60115-1, Clause 4.17** After immersing flux, dip in the 235±5℃ molten SAC solder bath for 2±0.5 sec.
Board Flex	2010/2512/1218： ΔR≦±(1%＋0.05Ω) Other size： F:ΔR≦±(0.5%＋0.1Ω) J:ΔR≦±(1%＋0.1Ω) No mechanical damage.	**IEC 60115-1, Clause 4.33** Resistance change after bended on the 90mm PCB, 10sec. 2512/2010/1218:2mm, 0402:5mm, else 3mm.
Short Time Overload	2010/2512/1218： ΔR≦±(2%＋0.1Ω) Other size： F:ΔR≦±(1%＋0.1Ω) J:ΔR≦±(2%＋0.1Ω)	**IEC 60115-1, Clause 4.13** 2.5 × Rated voltage for 5 seconds.

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Temperature Cycle	2010/2512/1218： ΔR≦±(1%＋0.05Ω) Other size： F:ΔR≦±(0.5%＋0.1Ω) J:ΔR≦±(1%＋0.1Ω)	**IEC 60115-1, Clause 4.19** Repeat 5 cycles as follow. -55℃(30min)→25℃(2~3min)→155℃(30min) →25℃(2~3min)
Load Life Humidity	2010/2512/1218： ΔR≦±(3%＋0.1Ω) Other size： F:ΔR≦±(1%＋0.1Ω) J:ΔR≦±(2%＋0.1Ω)	**IEC 60115-1, Clause 4.24** 40±2℃ with relative humidity 90% ~ 95% D.C. rated voltage for 1.5 hours ON 30 minutes OFF. Cycle repeated 1000 hours.

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Test Item	Specification	**Test Method**
Temperature Coefficient of Resistance (TCR)	Within the spec.	**IEC 60115-1, Clause 4.8** T_1 T_2 Test temperature： 25℃~ -55℃ 25℃~ +155℃ TCR(ppm/℃) = $(R_2-R_1)/R_1 \times 1 / (T_2-T_1) \times 10^6$
Load Life	2010/2512/1218： ΔR≦±(3%＋0.1Ω) Other size： F:ΔR≦±(1%＋0.1Ω) J:ΔR≦±(2%＋0.1Ω)	**IEC 60115-1, Clause 4.25** Rated voltage for 1.5 hours for followed by a pause 0.5 hour at 70±2℃. Cycle repeated 1000 hours.
Insulation Resistance	Between termination and coating must over 1000MΩ	**IEC 60115-1, Clause 4.6** Test voltage：100±15V

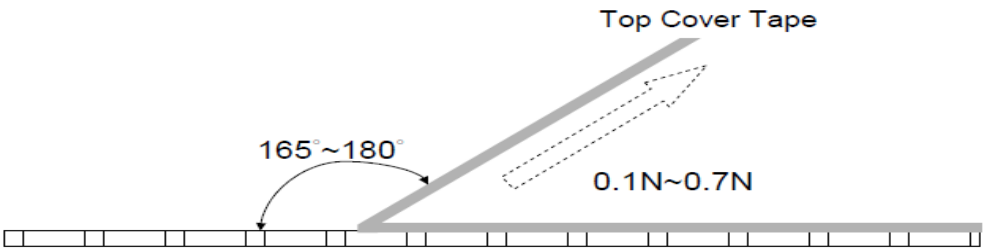
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9. PACKAGING

9.1 Peel Strength of Top Cover Tape

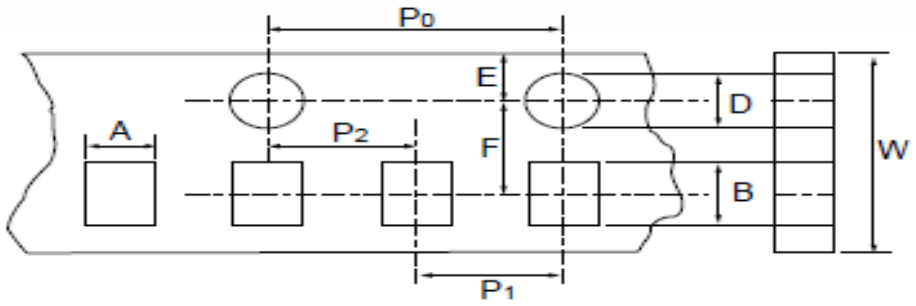
The peel speed shall be about 300 mm/min

The peel force of top cover tape shall between 0.1 to 0.7N



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9.2 Tape Packaging Dimensions



- Accumulated dimensional tolerance 40±0.2mm

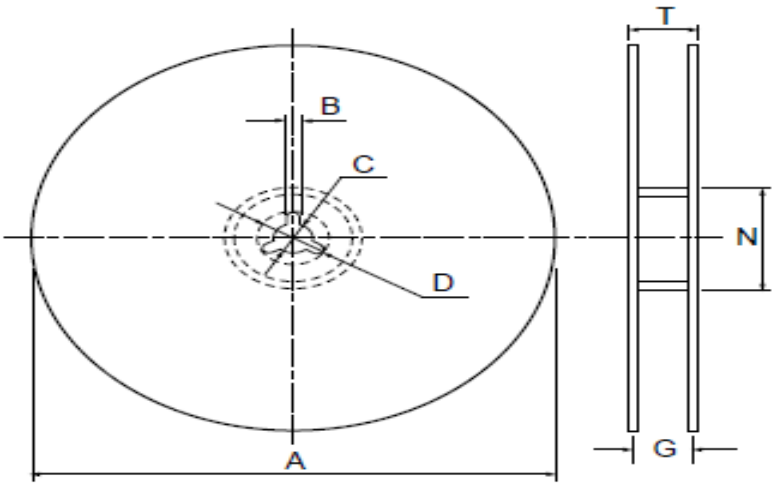
unit:mm

Size	A	B	W	F	E	P1	P2	P0	D
0402	0.70±0.10	1.20±0.10	8.00±0.30	3.50±0.20	1.75±0.10	2.00±0.10	2.00±0.05	4.00±0.10	1.50+0.10/-0
0603	1.10±0.20	1.90±0.20	8.00±0.30	3.50±0.20	1.75±0.10	4.00±0.10	2.00±0.05	4.00±0.10	1.50+0.10/-0
0805	1.65±0.20	2.40±0.20	8.00±0.30	3.50±0.20	1.75±0.10	4.00±0.10	2.00±0.05	4.00±0.10	1.50+0.10/-0
1206	2.00±0.20	3.60±0.20	8.00±0.30	3.50±0.20	1.75±0.10	4.00±0.10	2.00±0.05	4.00±0.10	1.50+0.10/-0
1210	3.00±0.20	3.60±0.20	8.00±0.30	3.50±0.20	1.75±0.10	4.00±0.10	2.00±0.05	4.00±0.10	1.50+0.10/-0
2010	2.80±0.20	5.50±0.20	12.00±0.30	5.50±0.10	1.75±0.10	4.00±0.10	2.00±0.05	4.00±0.10	1.50+0.10/-0
2512	3.50±0.20	6.90±0.20	12.00±0.30	5.50±0.10	1.75±0.10	4.00±0.10	2.00±0.05	4.00±0.10	1.50+0.10/-0
1218	3.55±0.30	4.90±0.20	12.00±0.30	5.50±0.10	1.75±0.10	8.00±0.10	2.00±0.05	4.00±0.10	1.50+0.10/-0

unit：mm

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9.3 Reel Dimensions



unit:mm

Size	Packaging Q'ty	A	N	C	D	B	G	T
0402	10kpcs/Reel	254.0±2.0	100.0±1.0	13.5±0.5	20(Min.)	2.0±0.5	10.0±1.5	14.9max.
	20kpcs/Reel	330.0±2.0	100.0±1.0	13.5±0.5	20(Min.)	2.0±0.5	10.0±1.5	14.9max.
0603 0805 1206 1210	5kpcs/Reel	178.0±2.0	60.0±0.5	13.0±0.5	20(Min.)	2.0±0.5	10.0±1.5	14.9max.
	10kpcs/Reel	254.0±2.0	100.0±1.0	13.5±0.5	20(Min.)	2.0±0.5	10.0±1.5	14.9max.
	20kpcs/Reel	330.0±2.0	100.0±1.0	13.5±0.5	20(Min.)	2.0±0.5	10.0±1.5	14.9max.
2010 2512	4kpcs/Reel	178.0±2.0	60.0±0.5	13.0±0.5	20(Min.)	2.0±0.5	13.8±1.5	16.7max.
1218	3kpcs/Reel	178.0±2.0	60.0±0.5	13.0±0.5	20(Min.)	2.0±0.5	13.8±1.5	16.7max.

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10. Storage &. Handling

… Products are recommended to be used up within one year as ensured shelf life.

Check solder ability in case shelf life extension is needed.

… To store products with following condition:

Temperature:5 to 40℃；Humidity: 20 to 70% relative humidity.

※ All product specification and data are subject to change without notice.

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